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Serum *N*-Glycan Profiling Identifies Candidate Glycan Biomarkers for Early Detection and Prediction of Alzheimer's Disease

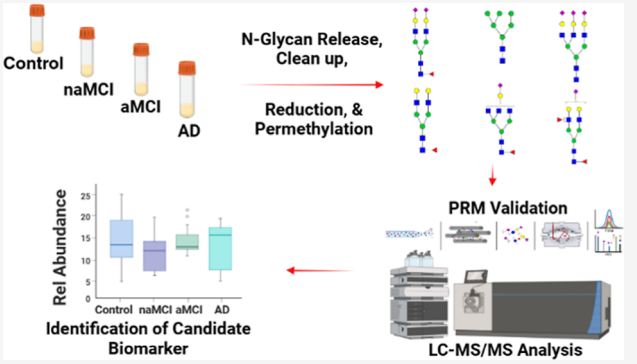
Sherifdeen Onigbinde, Joy Solomon, Cristian D. Gutierrez-Reyes, Oluwatosin Daramola, Mojibola Fowowe, Moyinoluwa Adeniyi, Kelly N. DuBois, Kelly M. Bakulski, Nicholas M. Kanaan, David M. Lubman, and Yehia Mechref*

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ABSTRACT: Alzheimer's disease (AD) is a neurodegenerative disorder marked by progressive cognitive decline, affecting millions worldwide. Early diagnosis and intervention are crucial yet challenging, particularly in distinguishing between Mild Cognitive Impairment (MCI) subtypes, which often precede AD. Recent studies have highlighted the significant role of glycosylation, specifically *N*-glycan alterations, in the pathogenesis of AD. This study utilizes advanced LC−MS/MS techniques to profile *N*-glycan changes between serum samples from participants with normal cognition denoted as (CTRL), nonamnestic MCI (naMCI), amnestic MCI (aMCI), and AD. The analysis identified 99 unique *N*-glycans and revealed distinct glycan expression patterns among the groups. Notably, sialylation was significantly upregulated in AD, while fucosylation was downregulated, suggesting their involvement in AD pathology. Additionally, the study examined the role of isomeric *N*-glycans, identifying several isomers that differentiate naMCI from aMCI and those that can monitor progression from aMCI to AD dementia. These findings emphasize the potential of *N*-glycans as biomarkers for early detection of AD and its precursors, offering new avenues for therapeutic intervention. The differential expression of specific *N*-glycans and their isomers could serve as valuable biomarkers for distinguishing MCI subtypes and predicting progression to AD dementia, thereby aiding in the development of targeted therapies aimed at mitigating cognitive decline in affected individuals.



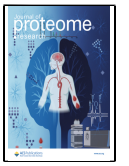
KEYWORDS: *Alzheimer's disease, mild cognitive impairments, biomarkers, N-glycans, isomers, LC-MS/MS*

INTRODUCTION

Alzheimer's disease (AD) is a progressive neurodegenerative disorder characterized by a gradual deterioration in memory, cognitive function, learning ability, and organizational skills.[1−3] It is the leading cause of dementia, impacting approximately 55 million individuals globally. The number of people affected by dementia is projected to surge to 140 million by 2050, highlighting the urgent necessity to address this escalating public health challenge.[2,4] Despite recent progress, significant challenges persist in the early detection and development of therapies to tackle AD. Research on early intervention and prevention requires a robust set of biomarkers that accurately diagnose early prodromal phases of AD, which is an area of significant effort in the field. Currently, clinical diagnosis of AD is based only on behavioral factors after all other potential causes have been ruled out. Blood-based biomarkers have great potential to improve access to early risk assessments and provide a faster platform for clinical trials in AD. Annually, about 10−15% of Mild Cognitive Impairment (MCI) patients progress to AD dementia[5] thus, it is essential to identify the elements contributing to the transition from MCI to AD in blood samples as a possible means for early intervention and prevention of AD.

MCI signifies a gradual shift from normal cognitive functioning to dementia and characterizes the early onset of memory loss.[6−8] People with MCI have higher levels of cognitive decline compared to others in the same age group. Nevertheless, they retain the ability to function autonomously and, as a result, cannot be diagnosed as demented.[9] Because of the heterogeneity associated with the clinical outcomes of MCI patients, they have been further categorized into two phenotypes: amnestic MCI (aMCI) and nonamnestic MCI (naMCI). Individuals with aMCI experience memory loss as the primary symptom, which is strongly linked to a significant

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Table 1. Serum Sample Clinical Information[a]

serum samples	control	naMCI	aMCI	AD
number	28	10	14	12
sex (male/female)	11/17	1/9	6/8	6/6
age in years (mean ± SD)	74 ± 7	71 ± 4	79 ± 8	71 ± 7
diabetes (yes/no)	0/28	0/10	0/14	1/11
cancer (yes/no)	0/28	0/10	0/14	0/11
pT181 (mean ± SD)	2.4 ± 1.1	1.7 ± 1.3	2.7 ± 1.5	4.9 ± 1.7
pT271 (mean ± SD)	0.4 ± 0.2	0.3 ± 0.1	0.7 ± 0.5	1.3 ± 0.4
Tau (mean ± SD)	3.7 ± 1.2	3.5 ± 1.1	3.7 ± 1.1	4.3 ± 2.5
aBeta42 (mean ± SD)	8.3 ± 2.0	6.8 ± 1.6	7.5 ± 1.8	6.9 ± 1.8
aBeta40 (mean ± SD)	206.9 ± 38.5	158.1 ± 37.8	204.4 ± 57.8	185.9 ± 47.2
aBeta42/40 (mean ± SD)	0.04 ± 0.01	0.04 ± 0.01	0.04 ± 0.01	0.037 ± 0.002
GFAP (mean ± SD)	154.6 ± 80.2	131.7 ± 44.9	217.0 ± 137.6	325.9 ± 139.2
NfL (mean ± SD)	18.0 ± 6.8	13.0 ± 4.0	18.6 ± 9.2	26.4 ± 9.9

[a]naMCI: nonamnestic mild cognitive impairment, aMCI: amnestic mild cognitive impairment, AD: Alzheimer's disease.

likelihood of progressing to AD, while naMCI is associated with impairment in one or more cognitive areas other than memory and often advances to non-Alzheimer's dementia.[10−13] This information is important for making accurate clinical predictions and assessing the risk level of individuals, which can help guide counseling and determine appropriate treatment options.

Recent research indicates that glycosylation plays a significant role in AD.[14,15] Given that many molecules associated with AD are either glycan-modified or involved in glycan regulation,[14,16] exploring glycobiology offers a novel perspective on understanding AD. This insight presents promising avenues for developing new therapeutic strategies targeting glycan-related mechanisms in the disease. Glycosylation is an important regulatory process that controls protein folding, molecular transportation, cell adhesion, receptor activation, and signal transmission.[17,18] Aberrant glycosylation of proteins has been linked to various diseases, including cancers[19−25] bacterial/viral infections,[26−28] MCI,[29,30] and AD.[29−35]

The current understanding of the role of glycosylation in AD suggests a pivotal involvement of glycan modifications in disease pathology.[31,34,36,37] These glycosylation changes may occur early during the development of AD, even before clinical symptoms manifest, and can involve alterations in the glycan structures attached to proteins implicated in AD, such as amyloid-beta (Aβ)[37−39] and tau.[40,41] Before the AD dementia stage, patients transition from normal cognition to MCI, characterized by subtle cognitive changes. Research indicates that specific glycan modifications on proteins could serve as biomarkers for detecting MCI and predicting progression to AD. For instance, aberrant glycosylation patterns on cerebrospinal fluid (CSF) and serum proteins have been associated with MCI and AD,[42−44] highlighting a potential link between glycosylation changes and early cognitive decline. Despite significant progress, critical gaps persist regarding how glycosylation changes correlate with different stages of cognitive decline. To the best of our knowledge, there is no study showing alterations in glycosylation, specifically the *N*-glycomics changes from normal cognition to naMCI, aMCI, and AD dementia. As a pressing need remains to comprehensively elucidate temporal correlations between glycosylation alterations and AD progression stages, understanding how the changes in *N*-glycans correlate with the different stages of cognitive decline could identify potential *N*-glycan biomarkers for the early diagnosis of AD, predict AD progression, and inform targeted therapeutic interventions.

Liquid chromatography−mass spectrometry (LC−MS/MS) is a reliable and sensitive analytical tool for glycomics studies because of its high sensitivity and specificity.[45−48] The high level of sensitivity enables the detection and measurement of glycans, even at very low concentrations, within intricate biological samples such as serum and CSF.[49] Therefore, LC−MS/MS *N*-glycomics approaches have been adopted in the study of several diseases such as cancer,[20,50] diabetes,[51] and neurodegenerative disorders like AD.[52,53] The ability of LC−MS/MS to precisely identify and measure glycans at low concentrations in diverse biological matrices such as serum has revolutionized our understanding of disease mechanisms and biomarker discovery. This analytical technique provides valuable insights into the molecular intricacies of these conditions and holds promise for developing novel diagnostic and therapeutic strategies.

In this study, we analyzed the serum *N*-glycome changes associated with the development and progression of AD dementia from healthy control, naMCI, and aMCI by employing the highly sensitive LC−MS/MS technique. We utilized a precise parallel reaction monitoring (PRM)-MS method to validate the changes in the expression of the identified *N*-glycans. Additionally, we profiled the isomeric *N*-glycans to see the contribution and roles of *N*-glycan isomers in the development of the disease. This study will be the first to examine *N*-glycan isomerism in AD and provide a detailed profiling of *N*-glycans in MCI subtypes.

MATERIALS AND METHODS

Sample Information

Human blood serum samples were obtained from the Michigan Alzheimer's Disease Research Center (MADRC), a collaboration of the University of Michigan, Michigan State University, and Wayne State University, to study various aspects of AD through the University of Michigan Memory and Aging Project (UM MAP). Participant data collection followed the Uniform Data set version 3 (UDSv3) from the National Alzheimer's Coordinating Center (NACC). The closest blood draw to participants' clinical consensus diagnosis was used and ranged from 0 to 143 days (M days = 14.70, SD = 31.50) and the serum samples were collected between 2019 and 2023. This study included samples from participants: 10

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with naMCI, 14 with aMCI, 12 with AD, and 28 with normal cognition. Clinical consensus diagnoses (among three clinicians and informed by neuropsychiatric tests) for the participants were made according to guidelines from the National Institute on Aging and the Alzheimer's Association (NIA-AA).[10] For the control group, participants were selected based on the absence of significant conditions, including cancer, diabetes, neurological diseases, severe inflammation, major head trauma, and alcoholism. Participants provided written, informed consent at the time of participation. The MADRC was approved by the University of Michigan Institutional Review Board (HUM00000382). For this study, serum samples were all deidentified and obtained from the MADRC archives. A summary of the clinical data (age, sex, diabetes, status) related to the samples used in this study can be found in Table 1.

Chemicals and Reagents

Ammonium bicarbonate (ABC), borane-ammonia, acetic acid, formic acid (FA), dimethyl sulfoxide (DMSO), iodomethane, sodium hydroxide (NaOH) beads, and mesoporous graphitized carbon (MGC) materials were acquired from Sigma-Aldrich (St. Louis, MO, USA). PNGase F enzyme was obtained from New England Biolabs (Ipswich, MA, USA), and difluoroacetic acid (DFA) was obtained from Acros Organics (New Jersey). Isopropyl alcohol (IPA), acetonitrile (ACN), methanol (MeOH), and water of HPLC grade were obtained from Fisher Scientific (Fair Lawn, New Jersey, USA). Microcolumns were purchased from Harvard Apparatus (Holliston, MA, USA), and the Isolute C18 (EC) cartridges were obtained from Biotage (Charlotte, NC, USA). The fused silica capillary was obtained from Polymicro Technologies located in Phoenix, Arizona. The Kasil frit kit was purchased from Next Advance in Troy, New York.

Release and Purification of *N*-Glycans

Initially, a protein assay was conducted using a Micro BCA protein assay kit (Thermo Sci, Rockford, IL), and volumes equivalent to 20 μg of protein from the assay were transferred into Eppendorf tubes. The samples were diluted to a final volume of 50 μL using a 50 mM ABC buffer. The mixture was then heated to 90 °C for 15 min to denature the proteins. Following this, 1000 U of PNGase F was added to each sample and then incubated at 37 °C for 18 h to facilitate efficient enzymatic digestion. After digestion, the samples were dried in a vacuum concentrator and then reconstituted in 300 μL of 5% acetic acid to eliminate deglycosylated proteins from the mixture using a previously described C18 SPE cleanup method. Briefly, SPE C18 cartridges were prepared by rinsing with 1 mL of methanol three times. This was followed by equilibrating them with 1 mL of a 5% acetic acid solution three times. Subsequently, the reconstituted samples were placed onto the cartridges and subjected to three washes using 300 μL of 5% acetic acid. The eluent from this process was collected in 1.5 mL tubes and evaporated using a vacuum concentrator.

Reduction and Permethylation of *N*-Glycans

The released and purified *N*-glycans were reduced following a protocol from a previously published method.[42] Briefly, a solution of borane-ammonia complex was freshly prepared at a concentration of 10 μg/μL in HPLC-grade water. To each sample, 10 μL of fresh borane-ammonia complex solution was added and incubated at 60 °C for 1 h. Afterward, the excess borane-ammonia in the samples was removed by repeatedly adding 1000 μL of methanol, until the resultant methyl borate was fully evaporated using a vacuum concentrator. Following reduction, the *N*-glycans were subjected to solid-phase permethylation using an established protocol.[54,55] This process involved reconstituting the reduced *N*-glycan samples in 30 μL of DMSO, adding 1.2 μL of water, and 20 μL of iodomethane. Microspin columns initially packed with NaOH beads suspended in DMSO were spun at 1800 rpm for 2 min, then washed with 200 μL of DMSO and spun again at the same speed. The sample solution was transferred into the columns and then incubated in the dark at room temperature for 25 min. An additional 20 μL of iodomethane was added, and the column was incubated for 15 min more. Following this incubation, the columns were spun at 1800 rpm for 2 min to collect the eluent. Lastly, 30 μL of ACN was added for a second elution. The permethylated *N*-glycans were then collected by centrifugation, dried, and reconstituted in a 20% acetonitrile and 0.1% formic acid aqueous solution for LC−MS analysis.

LC−MS/MS Analysis

150 mm C18 Column LC−MS/MS Conditions. We employed our previously established protocol[30,56−58] for the LC−MS/MS analysis of *N*-glycans released from serum samples of 12 AD, 10 naMCI, 14 aMCI, and 28 healthy control participants. Volumes of reduced and permethylated *N*-glycans, equivalent to the amount derived from 1 μg of protein, were injected into the LC−MS/MS system. Analysis was performed using an UltiMate 3000 Nano UHPLC system (Thermo Sci., San Jose, CA, USA) coupled with an Orbitrap Fusion Lumos mass spectrometer (Thermo Sci., San Jose, CA, USA) operated in a positive mode. For loading and online purification, an Acclaim PepMap 100 C18 trapping column was used (75 μm × 2 cm, 3 μm particle size, 100 Å pore size Thermo Sci., Pittsburgh, PA, USA) with a flow rate of 3 mL/min of mobile phase A (MPA) for 10 min. Chromatographic separation was subsequently achieved using a reversed-phase C18 Acclaim PepMap column (15 cm × 75 μm, Thermo Sci., Pittsburgh, PA, USA). The composition of MPA was 98% HPLC water with 2% ACN and 0.1% FA; mobile phase B was 98% ACN with 2% water and 0.1% FA. A chromatographic gradient with a column temperature of 55 °C and a flow rate of 0.35 mL/min was used. The chromatographic gradient started at 20% for 10 min, ramped to 55% in 35 min, and gradually increased to 90% in 5 min. Finally, it decreased to 20% and was kept constant for 5 min to equilibrate the column. After separation on the C18 column, the permethylated glycan samples were injected into the Orbitrap Fusion Lumos mass spectrometer. The spray voltage was set at 2 kV with a capillary temperature of 305 °C. The full scan was performed with a 120 K resolution, AGC target set to standard, an expected peak width of 30 s, the Max IT set to Auto, and a scan range of 400−2000 m/z. The MS/MS spectra were acquired in data-dependent mode for the top 20 most intense ions with a 30 K orbitrap resolution. Quadrupole was utilized for isolation with an isolation window of 2 m/z. A fixed CID collision energy of 35 was employed to fragment precursors. The AGC target was set to Standard and Max IT was set to Auto.

MGC LC−MS/MS Conditions for Isomeric Analysis

Utilizing an in-house packed 10 mm MGC column that has been demonstrated to provide efficient isomeric separation of permethylated glycans,[30,59] the *N*-glycan isomers involved with MCI and AD were profiled. For this analysis, an UltiMate 3000

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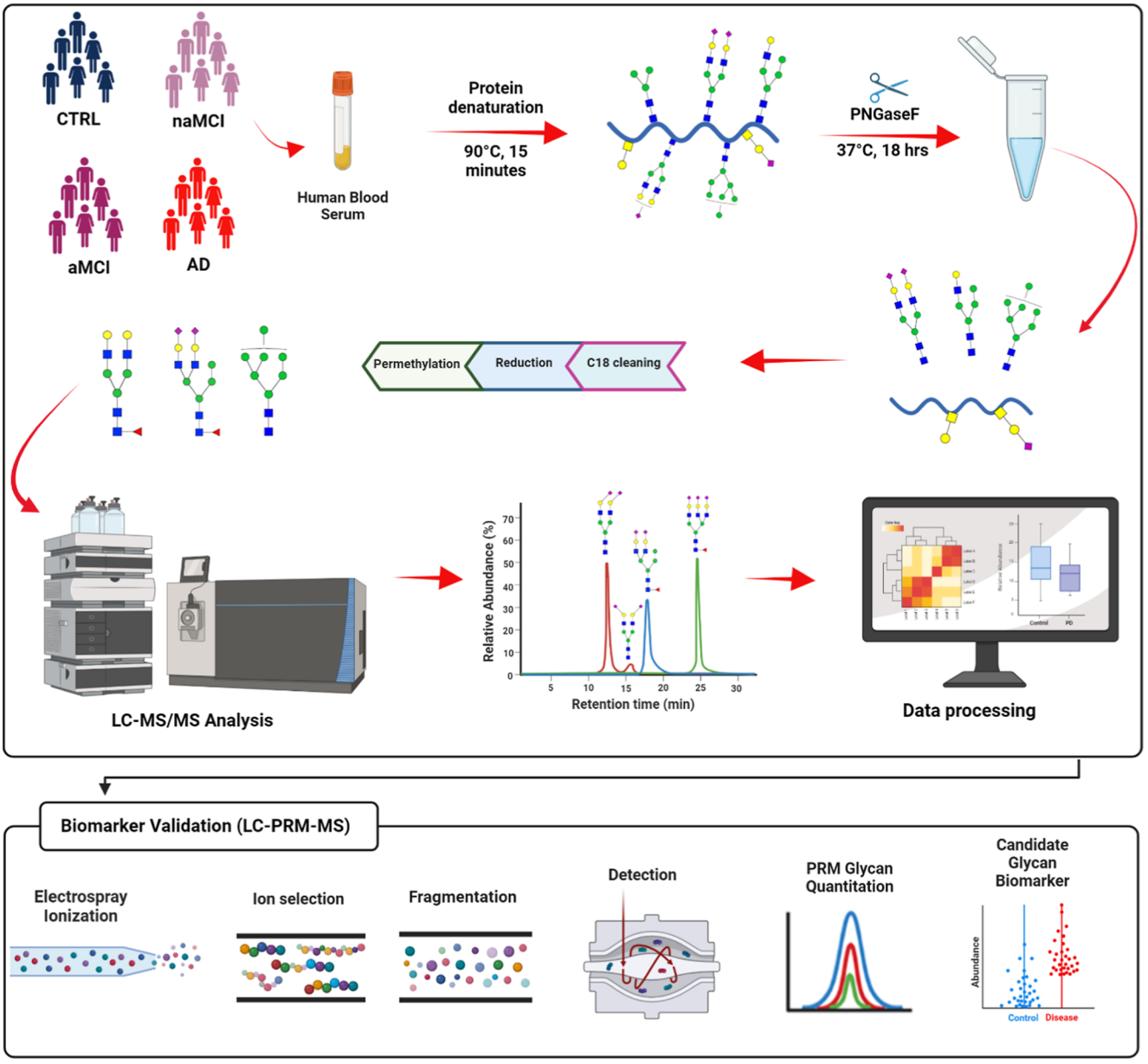


Figure 1. Schematics of the*N*-glycomics workflow employed for profiling serum *N*-glycans in this study. The serum samples were denatured followed by PNGase F digestion, reduction, and permethylation. The released *N*-glycans were analyzed utilizing advanced LC−MS/MS techniques. *N*-glycans were manually identified and validated using XCalibur software. Validation of *N*-glycan expression was performed using targeted PRM-LC-MS/MS technique. The *N*-glycan symbols used in this work include GlcNAc; Hex; Fucose; and NeuAc.

nano UHPLC system (Thermo Sci., San Jose, CA, USA) coupled with a Q-Exactive HF mass spectrometer (Thermo Sci., San Jose, CA, USA) operated in a positive mode was employed. Briefly, all samples were subjected to a 90 min multistep gradient on the MGC column for efficient separation of glycans. The MPA was composed of 98% water, 2% ACN, and 0.1% DFA, while MPB contained 50% ACN, 50% IPA, and 0.1% DFA. The column temperature was maintained at 75 °C and the flow rate was set at 0.3 μL/min. The gradient started at 20% MPB for 10 min, then increased to 60% over 20 min before it ramped to 95% over 30 min. It was kept constant at 95% for 20 min. The gradient was then reduced to 20% over 8 min and kept constant for 2 min to equilibrate the column.

After LC separation, *N*-glycans were analyzed using a mass spectrometer equipped with a nanoelectrospray ionization (ESI) source in positive ion mode. The spray voltage was maintained at 1.6 kV, and the transfer tube temperature was set to 275 °C. Full MS spectra were obtained using an Orbitrap mass analyzer, covering a mass range from 400 to 2000 m/z. The orbitrap resolution utilized is 120,000 and the accuracy is 5 ppm. The maximum injection time was set at 50 ms and the AGC target at 1×10^6.

The tandem MS/MS Orbitrap scan was performed in a data-dependent acquisition mode. The 20 most abundant precursor ions were selected for MS/MS scanning using a normalized collision energy (NCE) of 23%. The isolation window was set at 2 m/z and an AGC target of 1×10^5. The mass analyzer resolution was configured to 30,000, with a maximum injection time of 100 ms, and a loop count of 20.

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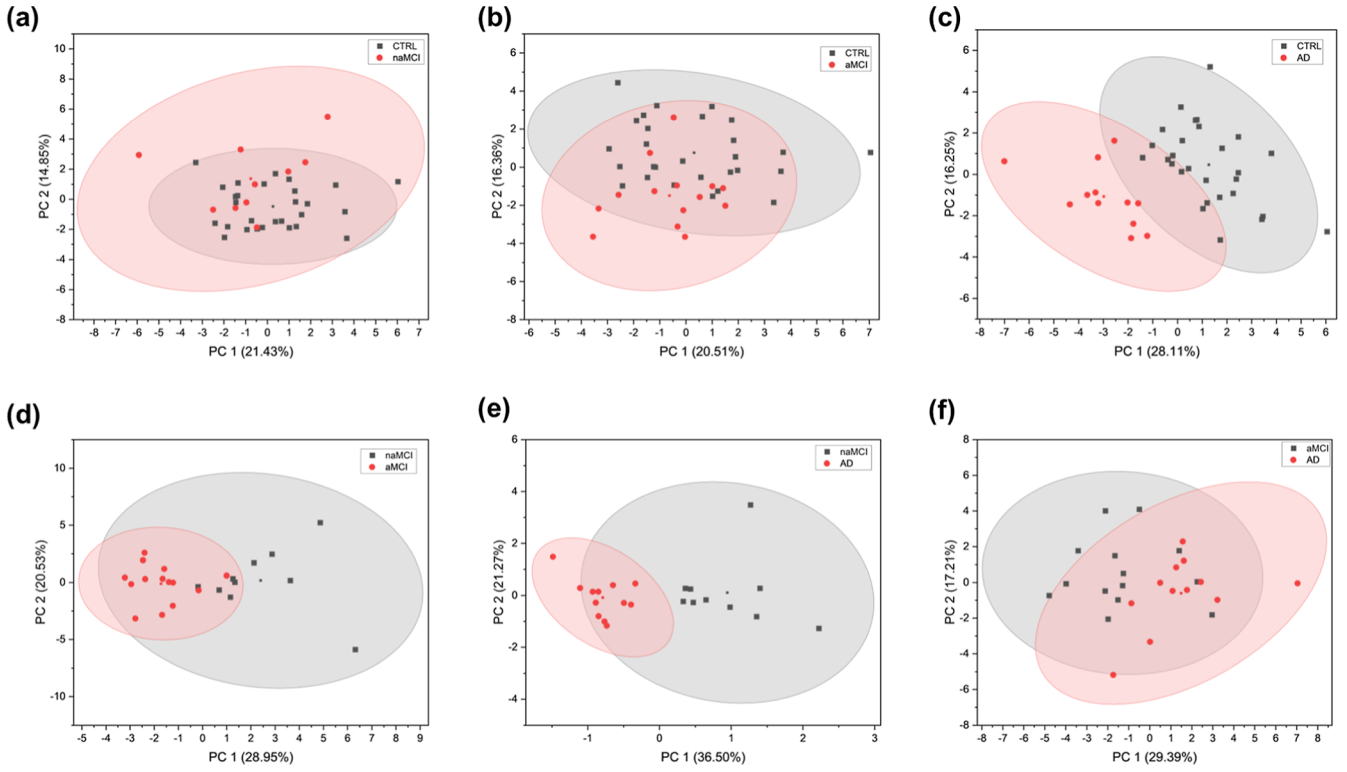


Figure 2. Quantitative changes in expression among the studied groups: Unsupervised Principal Component Analysis of *N*-glycans derived from Control, naMCI, aMCI, and AD groups at 95% confidence level. The PCAs show the comparison between (a) Control and naMCI, (b) Control and aMCI, (c) Control and AD, (d) naMCI and aMCI, (e) naMCI and AD, and (f) aMCI and AD. Symbols having the same color represent biological samples from the same group. Control (n = 28), naMCI (n = 10), aMCI (n = 14), and AD (n = 12).

LC-PRM-MS Condition

A targeted PRM strategy was utilized to validate the differential expression of *N*-glycans between the study groups. A transition list was created for the PRM analysis, which included key details about the *N*-glycans such as molecule names, precursor m/z, precursor charge, and transition fragment m/z as displayed in Table S1. The LC conditions are the same as described above for the C18 analysis. The PRM scan was performed with a 30k orbitrap resolution, the AGC target was set to standard, and Max IT was set to Auto. Quadrupole was utilized for isolation with an isolation window of 2 m/z. A fixed CID collision energy of 35 was employed to fragment targeted precursors. CID activation time was set to 10 ms while activation Q was 0.25. Quantitative validation was performed by analyzing PRM data using Skyline software (Version 21.2.0.536) for quantification. The normalized data were then analyzed statistically to compare PRM results across the various groups.

Data Analysis

The *N*-glycan compositions and structures were manually identified and validated using Xcalibur software (Version 4.2., Thermo Sci.). The retention time, monoisotopic mass, and MS/MS spectra were verified with a mass tolerance of 5 ppm. Due to the lack of diagnostic D ions in positive ion mode MS/MS, which are essential for confident arm-specific fragmentation, we were unable to confidently assign glycan branches such as the 3′ or 6′ arm. Consequently, ambiguous residues including sialic acids and fucose were denoted using brackets. Only *N*-glycans detected in at least 70% of the samples within each study group (CTRL, naMCI, aMCI, and AD) were included for analysis. No glycan included had a detection rate below this threshold in any group, and no missing value imputation was performed. The peak areas were quantified using Skyline software (Version 23.1.0.380) to represent the abundance of each *N*-glycan. Theoretical validation of the fragment ions was performed using Glycoworkbench (Version 2.0). The relative abundance of the *N*-glycans was computed in Microsoft Excel (Version 2308) by normalizing the peak area to the total abundance. Statistical analysis was carried out using IBM SPSS software (Version 29.0.1). The Mann−Whitney U test was employed, followed by the application of Benjamini-Hochberg's procedure to correct for multiple testing and control the false discovery rate. The Mann−Whitney U test was used to perform pairwise comparisons between specific independent groups, as it is a nonparametric method appropriate for non-normally distributed data without assuming homogeneity across all cohorts. IBM SPSS was also used to generate the receiver operating characteristic (ROC) curves. Origin software (Version 2.0) was employed for the unsupervised principal component analysis (PCA). Heatmaps were created using Genesis software (Version 1.8.1) while boxplots, volcano plots, and bar graphs were generated using GraphPad Prism (Version 10.2.2). Online Biorender software was employed to generate workflow schematics.

RESULTS AND DISCUSSION

N-Glycomics Workflow

The experimental workflow employed in this study is shown in Figure 1. For the experiment, serum proteins collected from the participants were denatured and digested using PNGase F to release *N*-glycans. A C18 cleanup step was then performed to remove the proteins. The purified glycans were subsequently

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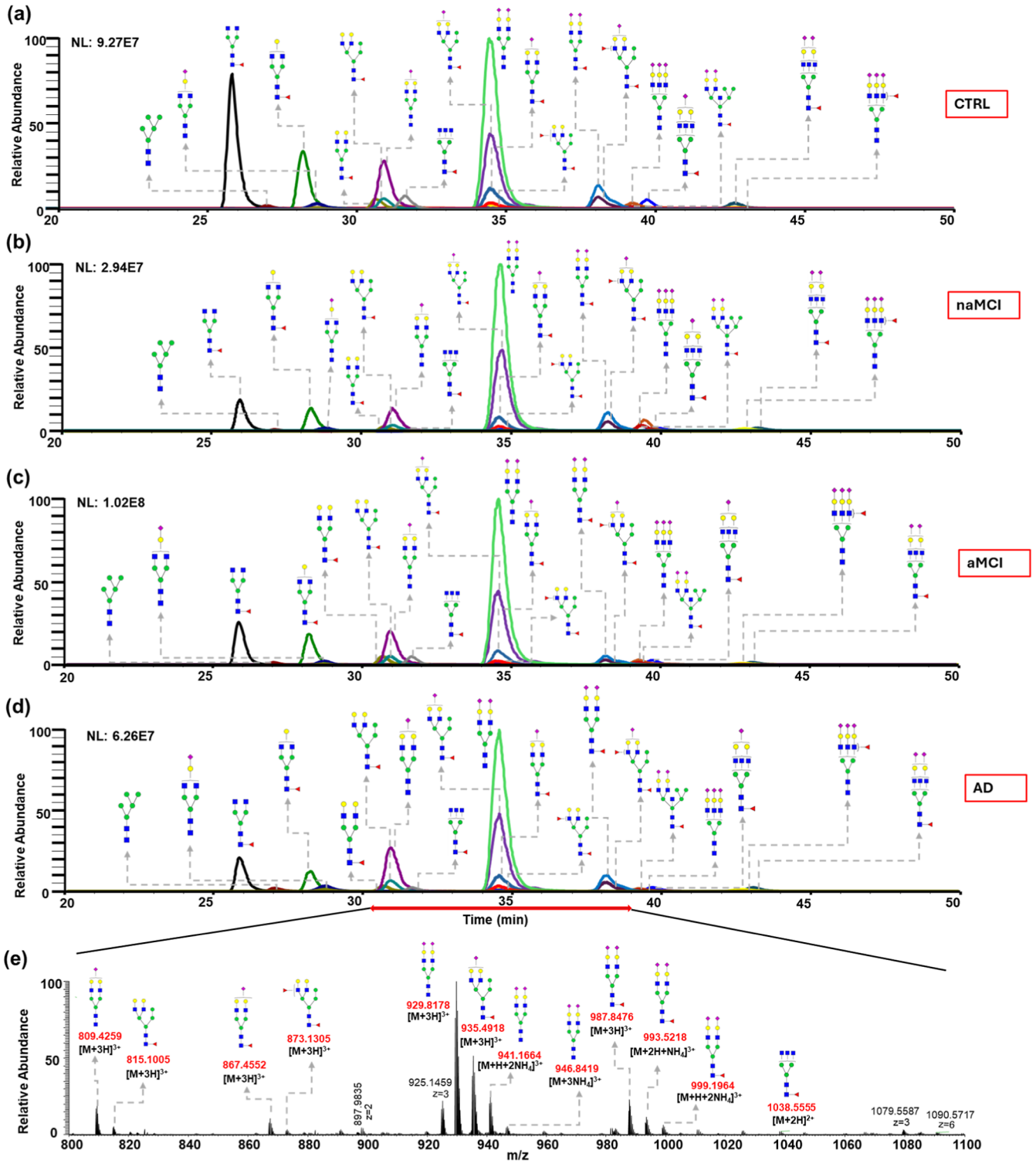


Figure 3. Extracted Ion Chromatograms (EICs) of permethylated*N*-glycans: The top four panels show the EICs of representative *N*-glycans in the four groups comparing their absolute expression in the following order: Control, naMCI, aMCI, and AD. The last panel represents the average mass spectra of the area highlighted in the red arrow.

reduced with borane ammonia complex and processed using a well-established solid-phase permethylation protocol.[54] This step made the glycans hydrophobic for RPLC analysis, stabilized them, enhanced MS detection, and prevented fucose rearrangement. Highly sensitive LC−MS/MS techniques were employed for the analysis of the released permethylated *N*-glycans from the 28 healthy controls, 10 naMCI,14 aMCI, and 12 AD samples. Then, the profiling of the *N*-glycans and their isomers derived from the serum samples were compared and investigated using various statistical tools. The chromatographic separation was initially achieved using a 150 mm long C-18 column in a 60 min run time to profile all the *N*-glycans in 64 serum samples. To analyze the influence of isomers on the changes associated with *N*-glycans in the control and

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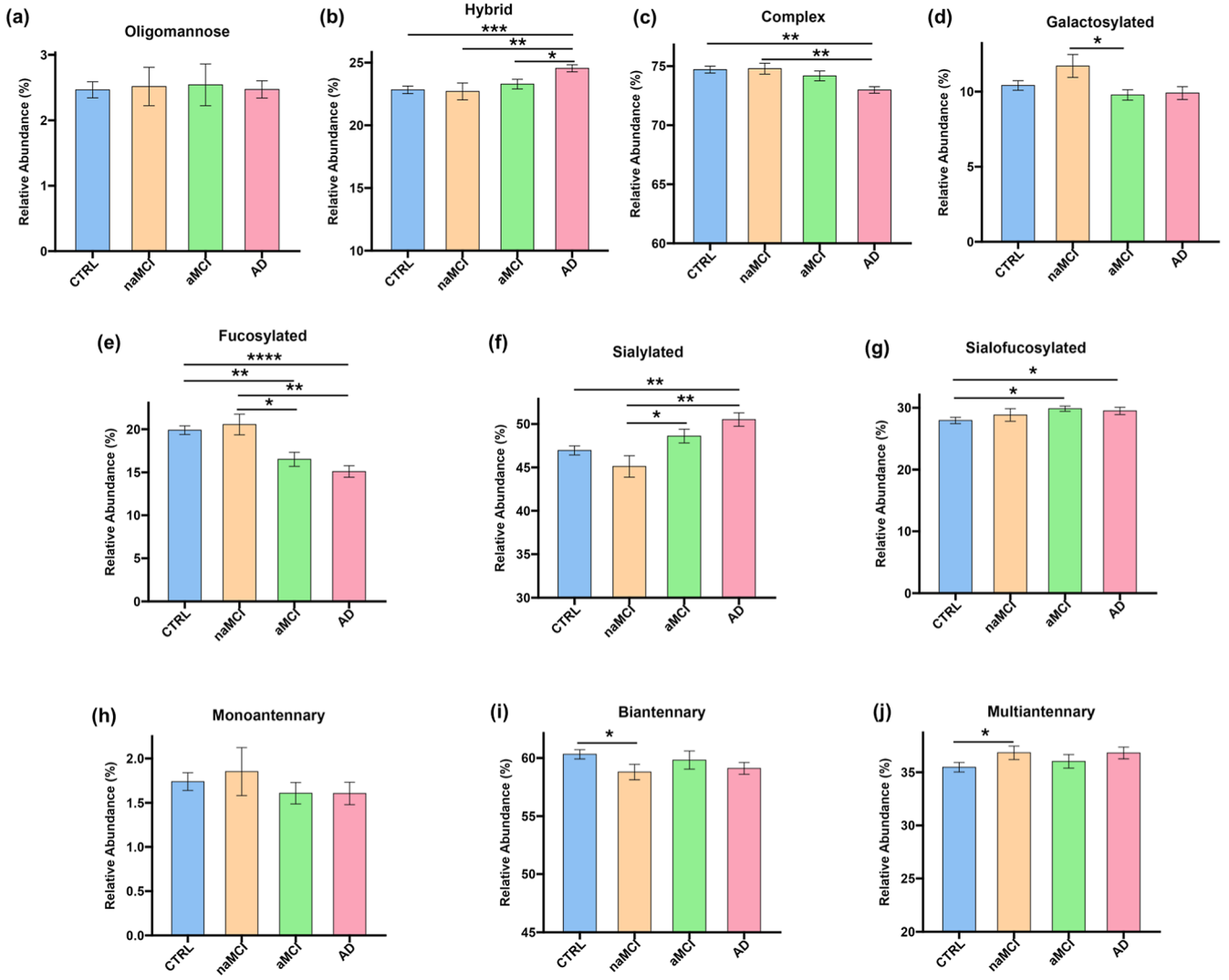


Figure 4. Distribution of*N*-glycan types among the four groups: The bar graphs represent the distribution of *N*-glycan types derived from the Control, naMCI, aMCI, and AD serum samples. The *N*-glycans are distributed according to their monosaccharide composition and configuration into the following categories: (a) oligomannose, (b) hybrid, (c) complex, (d) galactosylated, (e) fucosylated, (f) sialylated, (g) sialofucosylated, (h) monoantennary, (i) biantennary, and (j) multiantennary. Control (n = 28), naMCI (n = 10), aMCI (n = 14), and AD (n = 12). The error bars represent the standard error of mean (S.E.M). (* represents p-value < 0.05, ** represents p-value < 0.01).

disease groups, a 10 mm in-house packed MGC column was utilized. The *N*-glycan structures were annotated in the remaining part of the study using a four-digit nomenclature. For example, a glycan may be represented as "4512", where each digit corresponds to the number of monosaccharide units linked to a specific *N*-glycan. The numbering is in the following order: *N*-acetylglucosamine, Hexose, Fucose, and *N*-acetylneuraminic acid (HexNAc, Hex, Fuc, NeuAc). The symbols used to present the *N*-glycan in this work are shown in the caption of Figure 1.

Differential Expression of *N*-Glycans among the Sample Cohorts

A total of 99 *N*-glycans were identified across the analyzed serum samples utilizing the C18-LC-MS method. Table S2 provides a comprehensive list of identified glycans, including their nomenclature, glycan compositions (cartoons), theoretical m/z values, and corresponding mass accuracies. In addition to the *N*-glycans identified across all the studied groups, we also observed a subset of *N*-glycan compositions uniquely detected in individual groups. In the control group, glycan compositions 6720, 4531, 6601, and 3521 were exclusively detected. Glycans 5630 and 6640 were unique to the naMCI group, while 5800 and 3420 were identified only in the aMCI group. No unique glycans were observed in the AD group. These uniquely detected glycans were not detected in all other groups analyzed and may contribute to distinguishing group-specific glycomic profiles. Table S3 shows the list of the unique *N*-glycans detected in the individual groups. An unsupervised Principal Component Analysis (PCA) was employed to depict the quantitative differences in the data obtained from the investigated groups as shown in Figure 2. PCA is a mathematical approach that employs an orthogonal transformation to reduce complex data sets with high dimensions into lower-dimensional graphical plots that depict the relationships among data groups.[60] The discrepancy between the two data sets increases with increasing PCA distance. Using the calculated relative abundances of the identified glycan compositions, Figure 2 depicts the PCAs of the comparisons performed between each group. The clustering on the PCA indicates the discrepancies in glycan expression among the

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different groups analyzed. We observed the fewest changes in the glycan expression between the control group and those with naMCI as shown in Figure 2a, which may be due to the tendency of naMCI patients to recover cognitive function. In contrast, the glycan expression in aMCI participants was similar to AD participants (Figure 2f), likely because aMCI participants have a higher likelihood of progressing to AD. The most significant disparity was noted between the control group and AD as depicted in Figure 2c, followed by the naMCI and AD comparison (Figure 2e). The PCA in Figure 2d shows some discrepancy in the expression of *N*-glycans between the naMCI and aMCI subtypes of MCI. A combined PCA showing the relation among the four groups is shown in Figure S1. The changes in the expression of the individual *N*-glycans in these groups will be investigated in the later sections.

Figure 3 shows a representative trace of the extracted ion chromatogram of *N*-glycans identified in our analysis from the control, naMCI, aMCI, and AD groups. The last panel in this figure shows corresponding average mass spectra annotated with glycan structures. To evaluate the differences in overall *N*-glycan profiles among naMCI, aMCI, AD, and control groups, the expression levels of all identified *N*-glycans were relatively quantified, as shown in Table S4. Examples of tandem MS/MS spectra utilized for the characterization and assignment of *N*-glycans are shown in Figure S2. Examples including core and branch fucosylation, biantennary, multiantennary, complex, and hybrid *N*-glycan assignments are shown.

Glycosylation Characteristics and *N*-Glycan Types in the Sample Cohorts

Variation in the *N*-glycome profile among the groups was investigated by sorting identified *N*-glycans into different types according to their monosaccharide compositions and structures attached to the *N*-glycan core, which influence their biological functions and interactions. The relative abundance of *N*-glycans was calculated by normalizing the abundance of individual *N*-glycans to the total *N*-glycan abundance. Glycan types can be grouped in different ways to gain a deeper understanding of the diverse roles and structures of glycans in biological systems. In this work, the glycan types were defined as oligomannose, hybrid, complex, galactosylated, fucosylated, sialylated, sialofucosylated, monoantennary, biantennary, and multiantennary. Figure 4 describes the distribution of the percentage relative abundance of *N*-glycan types identified across the analyzed groups. No significant differences were detected in the levels of oligomannose or monoantennary glycans (Figure 4a,h, respectively). Analyzing the structures extending from the glycan core, complex-type *N*-glycans were found to be the most prevalent across all four groups, with an average relative abundance of 70% (Figure 4c). In the case of complex glycan types, a clear distinction could be made between the control and naMCI groups compared to the AD group, where a significant downregulation was observed. These findings suggest that this glycan type may be valuable for early detection of AD. For hybrid glycan types, a notable increase in expression was seen in the AD group compared to all other groups, including aMCI (Figure 4b). This glycan type could, therefore, be important for diagnosing AD and might serve as a predictive biomarker for its progression. In our analysis, the total abundance of galactosylated glycans was uniquely able to differentiate between the naMCI and aMCI groups (Figure 4d).

We observed a significant upregulation in sialylation in the AD group compared to all other groups except aMCI (Figure 4f). There was no significant difference in the expression of this *N*-glycan type between aMCI and AD. In the context of AD, altered sialylation of *N*-glycans has been associated with the pathology of the disease. This modification can influence the aggregation and clearance of Aβ plaques, which are a hallmark of AD.[61] Research has shown that increased sialylation leads to enhanced APP secretion and Aβ production indicating the role of glycosylation in AD pathogenesis.[39] Sialylated glycans are also known to play a role in modulating the immune response.[16,62−65] Blanchard and co-workers[66] also reported a significant upregulation of Biantennary sialylation in CSF of AD cases compared to controls. Our previous investigation comparing the *N*-glycan profiles of low-abundance glycoproteins in serum between controls and MCI cases demonstrated a notable elevation in sialylation and a reduction in fucosylation, consistent with our present observations.[30] Palmigiano et al.[42] reported increased bisected glycans and decreased sialylation in CSF of AD and MCI patients using MALDI-TOF-MS, but observed no significant changes in serum. Similarly, our previous CSF analysis in AD also showed elevated levels of bisected glycans and reduced sialylation, consistent with their findings.[33] In contrast, our current serum analysis revealed increased sialylation, decreased fucosylation, and distinct isomeric alterations in AD. These discrepancies may result from differences in analytical sensitivity, experimental design, or potential biological variability across cohorts. Our findings suggest that serum *N*-glycan profiling can reflect systemic signatures of AD pathology and may offer noninvasive biomarkers for early detection and monitoring of disease progression. In AD, chronic inflammation is a key feature. The upregulation of sialylation may affect the overall inflammatory environment in the brain, potentially contributing to neuronal damage and disease progression. Understanding the mechanisms underlying *N*-glycan sialylation upregulation in the serum of AD patients could reveal new therapeutic targets. In addition, modulating glycosylation enzymes or pathways involved in sialylation may offer novel approaches to slow or halt disease progression. Furthermore, *N*-glycan sialylation can impact the trafficking and degradation of glycoproteins, including those involved in clearing Aβ.[67] Enhanced sialylation might hinder these processes, contributing to the accumulation of toxic protein aggregates. Neurofibrillary tangles, a pathological hallmark of Alzheimer's disease, have been found to be hypersialylated, contributing significantly to the accumulation of phosphorylated tau (p-tau) in AD.[68]

Conversely, fucosylation was significantly downregulated in AD compared to all other groups except aMCI (Figure 4e). Although there is a decrease in the trend of fucosylation in AD compared to aMCI, this was not statistically significant. Altered fucosylation can reflect changes in immune responses and neuroinflammation, which are key features of AD. Gu and co-workers[69] have shown that core fucosylation regulates neuroinflammation and decreased expression of core fucosylation could increase the sensitivity of glial cells to inflammatory mediators. Sialofucosylation was increased in the comparison between the control and aMCI group as well as the AD group (Figure 4g). This may be due to the overall influence of sialylation on these types of glycan structures.

To further investigate the sialylation and fucosylation types that are influencing these major changes, we subdivided the

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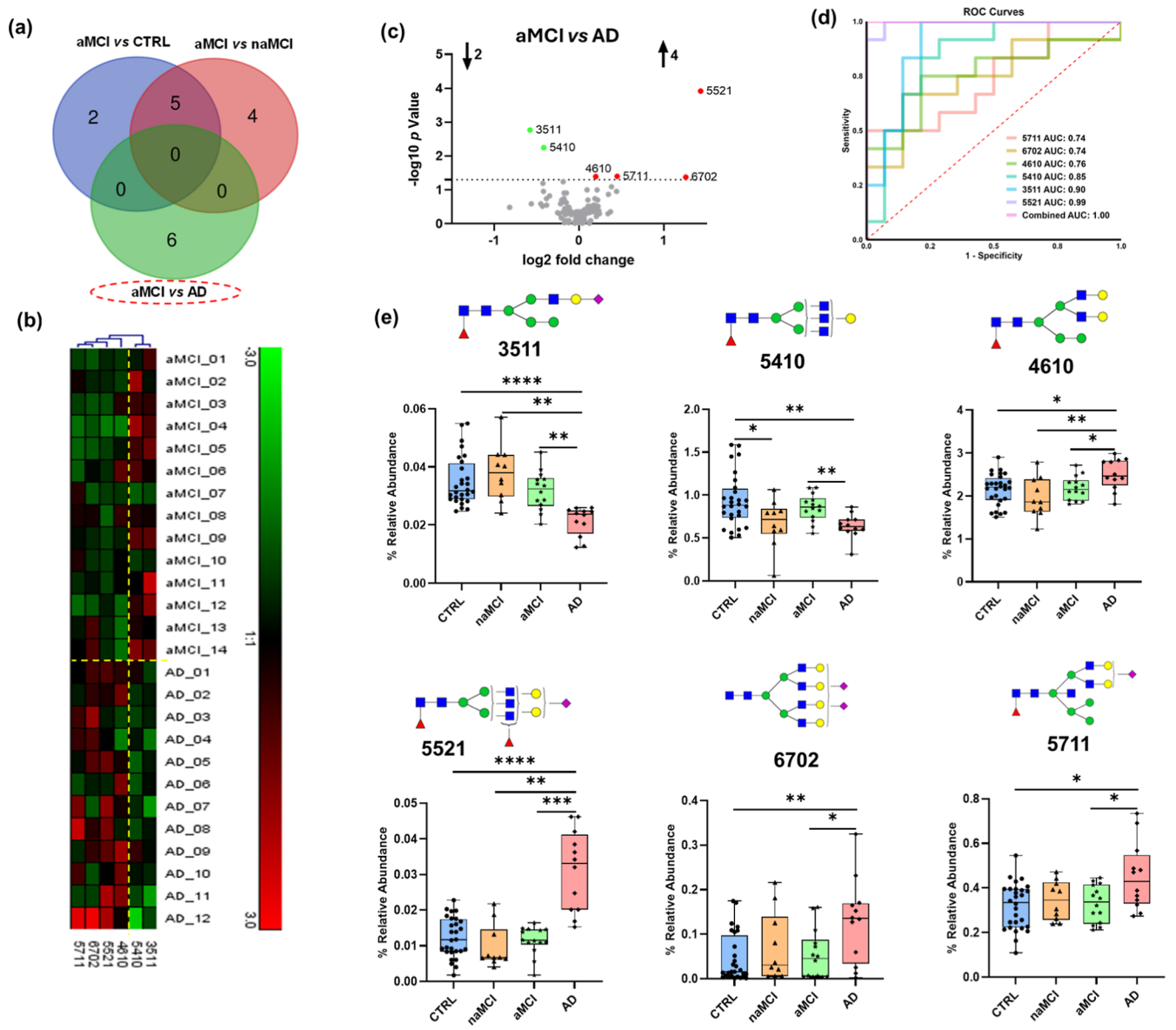


Figure 5. Variation in *N*-glycan expressions between amnestic MCI and AD: (a) the Venn diagram shows the unique and shared statistically significant *N*-glycans in aMCI compared to Control, naMCI, and AD; (b) the heatmap shows the *N*-glycans with significant changes in expression between aMCI and AD; (c) the volcano plot shows the distribution of *N*-glycans according to their expression in MCI subtypes; (d) the ROC curves show the sensitivity and specificity of significant *N*-glycans between aMCI and AD comparison; and (e) the box plots represent the percentage relative abundance of statistically significant *N*-glycans in the comparison between aMCI and AD. (* represents *p*-value < 0.05, ** represents *p*-value < 0.01). *N*-glycan nomenclature and composition as described in Figure 1.

sialylation and fucosylation into different types according to the number of sialic acid or fucose moieties attached to the *N*-glycan structures. Sialylated glycans were categorized into monosialylated, disialylated, trisialylated, and tetra-sialylated groups, while fucosylated glycans were divided into mono- and bifucosylated categories (Figure S3). This additional analysis showed that disialylation is the primary contributor to the observed shifts in glycan expression across the four study groups (Figure S3a). Trisialylation was linked to the downregulation found when comparing naMCI to aMCI. No significant changes were observed in the mono- and tetra-sialylated groups across all four categories. We also analyzed fucosylation based on the number of attached fucose molecules, classifying them into mono- and bifucosylated glycans. The notable reduction in fucosylation detected in AD and aMCI, compared to the control and naMCI groups, was primarily driven by monofucosylated glycans (Figure S3b). In Figure S3c, we have shown that core fucosylation was significantly decreased in aMCI and AD when compared to the control and naMCI groups. We investigated the MS/MS spectra of fucosylated glycans to determine core fucosylation. Figure S4 shows MS/MS spectra of representative core fucosylated glycans and the fragments indicative of their core fucosylation.

Additionally, the relative abundance of biantennary and multiantennary glycans helped distinguish the control group from the naMCI group (Figure 4i,j, respectively).

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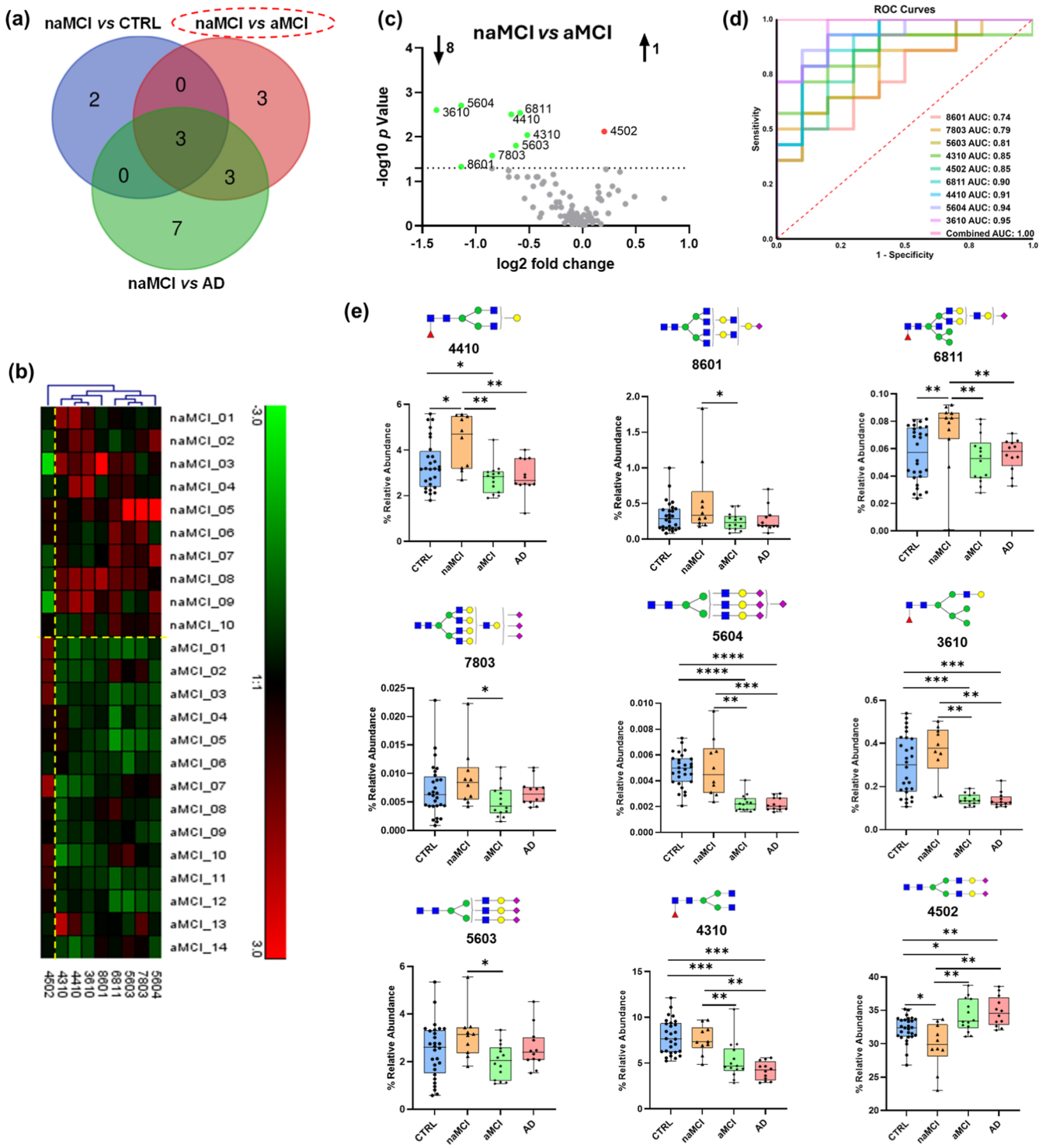


Figure 6. Distinguishing the subtypes of MCI: Glycan profiling differentiates naMCI from aMCI. (a) The Venn diagram depicts the overlap of statistically significant *N*-glycans in the comparison between naMCI and other groups; (b) the heatmap shows the *N*-glycans expression changes between naMCI and aMCI; (c) the volcano plot shows the distribution of *N*-glycans according to their expression in MCI subtypes; (d) the ROC curves showing the sensitivity and specificity of significant *N*-glycans between naMCI and aMCI comparison; and (e) box plots representing the percentage relative abundance of statistically significant *N*-glycans in the comparison between naMCI and aMCI. (* represents p value < 0.05, ** represents p value < 0.01). *N*-glycan nomenclature and composition as described in Figure 1.

N-Glycan Profiling Identifies Predictive Biomarkers of AD and Differentiates MCI Subtypes

MCI is a heterogeneous condition with diverse subtypes, risk factors, clinical presentations, and progression rates.[12,70] MCI has been previously defined into two primary subtypes: amnestic MCI (aMCI) and nonamnestic MCI (naMCI).[71] People with aMCI primarily experience memory loss, while those with naMCI show deficits in areas that include executive functions, attention, and language.[72] While aMCI is a distinct condition, it can be considered an intermediate stage between

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the cognitive decline associated with normal aging and the more severe decline seen in dementia, including AD.[10] In contrast, memory loss is more pronounced in AD, which is often accompanied by impairment in other cognitive functions such as language, problem-solving, and judgment. Research has indicated that aMCI reflects the earliest symptomatic stage of AD, whereas naMCI is more likely to lead to non-AD forms of dementia.[70,73] Understanding the differences in the glycome expression of aMCI and AD is crucial for early diagnosis and intervention, which can potentially slow the progression of cognitive decline.

Next, we compared the *N*-glycome profile in aMCI and AD to identify *N*-glycan biomarker candidates that can predict progression to AD because of the high conversion rate of aMCI patients to AD. We employed a Venn diagram to describe the unique and shared glycan compositions in the direct comparison of aMCI to control, naMCI, and AD (Figure 5a). Six different *N*-glycan compositions show differential expressions between aMCI and AD. Interestingly, the changes in the expression of this set of glycans were unique to only the aMCI and AD comparison. The heat map in Figure 5b depicts the expression of the significantly altered *N*-glycans in this comparison. The red color indicates the upregulation of *N*-glycans, while the green color represents downregulation. When comparing aMCI and AD, two *N*-glycans including 3511 and 5410 were downregulated in AD while four *N*-glycan compositions including 4610, 5521, 6702, and 5711 were upregulated. Figure 5c shows the volcano plot of the aMCI vs AD comparison. Due to the critical role of glycans as potential biomarkers, it is essential to have precise techniques for evaluating their diagnostic efficiency. ROC curves were applied to assess the diagnostic performance of differentially expressed *N*-glycans between aMCI and AD, providing insight into the reliability of these glycan expressions as biomarkers. In ROC curves, an Area Under the Curve (AUC) value closer to 1 indicates a better performance of the molecules to discriminate between groups while a value of 0.5 indicates no discrimination. *N*-glycans 5711 and 6702 show the lowest AUC value of 0.74 while *N*-glycan 5521 shows the highest AUC value of 0.99 between aMCI and AD groups (Figure 5d). Combined, they show an AUC value of 1, indicating that this panel of *N*-glycans can be an efficient diagnostic toolkit for monitoring the progression of aMCI to AD dementia to enable early intervention. This rivals the discriminatory power seen using plasma phosphorylated tau species as biomarkers to distinguish between Aβ PET-positive and -negative individuals.[74−77] This is particularly striking because the UM-MAP cohort is a community-based cohort that lacks the extensive biomarker characterization provided by more invasive and expensive procedures, such as PET and CSF biomarker analysis. The diagnostic utility of plasma AD biomarkers is often reduced in this context.[78] Additionally, phosphorylated tau species are often unable to segregate individuals with MCI from those with AD when individuals are not stratified by Aβ PET,[79−81] suggesting a potential unique utility for *N*-glycan analysis. The box plots of the significant *N*-glycans in this comparison appear in Figure 5e which shows the expression of these six *N*-glycans in the four different groups analyzed in this study.

Since MCI is heterogeneous, we also investigated the changes in the *N*-glycome profile of naMCI and aMCI to identify *N*-glycan structures that can differentiate these subtypes. The Venn diagram in Figure 6a shows that nine *N*-glycans in total can differentiate naMCI from aMCI. The heat map in Figure 6b depicts the expression of the significantly altered *N*-glycans between the naMCI and aMCI groups. Of the 9 total significant *N*-glycans, 8 structures including 5604, 7803, 5603, 6811, 8601, 3610, 4410, and 4310 were downregulated in aMCI compared to naMCI, while only *N*-glycan 4502 was upregulated in this comparison. The majority of the downregulated *N*-glycan structures are multiply sialylated and monofucosylated structures. A volcano plot that depicts the expression of all *N*-glycans in this comparison appears in Figure 6c. The individual glycans in this comparison have AUC values ranging between 0.74 to 0.95 and a combined AUC of 1.00 suggesting a good performance of these *N*-glycans for distinguishing naMCI and aMCI (Figure 6d). The box plots of the significant *N*-glycans in this comparison are shown in Figure 6e. The figure shows the expression of these 9 *N*-glycans in the four different groups analyzed in this study. As MCI is a heterologous condition with few clear mechanisms for differentiating between naMCI and aMCI, *N*-glycans may have utility as biomarkers of aMCI and may predict future progression to AD.[82,83] One of the significant glycan 5604 that decreased in aMCI and AD may be a polysialylated structure. Polysialylated *N*-glycans have been implicated in neuroinflammation, and their decrease in the serum of AD patients may reflect impaired systemic immunomodulation and altered leukocyte function, potentially contributing to or mirroring neuroinflammation.[84] Although the panel of candidate *N*-glycan biomarkers identified demonstrates strong discriminatory power for distinguishing MCI subtypes and predicting progression to AD, it is important to note that these glycans are relatively low in abundance. This characteristic may present challenges for clinical translation, particularly regarding detection sensitivity and assay reproducibility. Therefore, successful implementation in clinical settings would require highly sensitive analytical platforms and further validation studies in larger, independent cohorts to ensure robustness and reproducibility of the findings. Future longitudinal studies in large diverse cohorts will further elucidate the prognostic role of *N*-glycans. This may also provide important information in clinical trial stratification or decision making for therapeutic interventions such as anti-Aβ immunotherapies.

Comparisons between other groups including CTRL vs AD, CTRL vs naMCI, CTRL vs aMCI, and naMCI vs AD were also investigated. Figure S5 shows the heat maps of the statistically significant *N*-glycans in the four comparisons. The volcano plots depicting the expression of all glycans in these comparisons are shown in Figure S6. Box plots of the remaining significant *N*-glycans in any of the four comparisons are shown in Figure S7. The ROC curves showing the performance of these *N*-glycan structures as potential candidate biomarkers for AD diagnosis are shown in Figure S8. The Venn diagram showing the comparison of *N*-glycans identified in other groups compared to control is shown in Figure S9a, while the comparison between other groups and naMCI is shown in Figure S9b.

In the recently revised criteria for the diagnosis and staging of AD, there is a minimum diagnostic accuracy requirement (90% or greater) for diagnostic biomarkers in the target population.[85] It is worth noting that in our study, *N*-glycans meet this stringent criterion and rival the diagnostic capabilities of the widely accepted assays detecting plasma phosphorylated tau species.[86,87] Indeed, especially in individuals with mild cognitive impairment, phosphorylated tau species are currently

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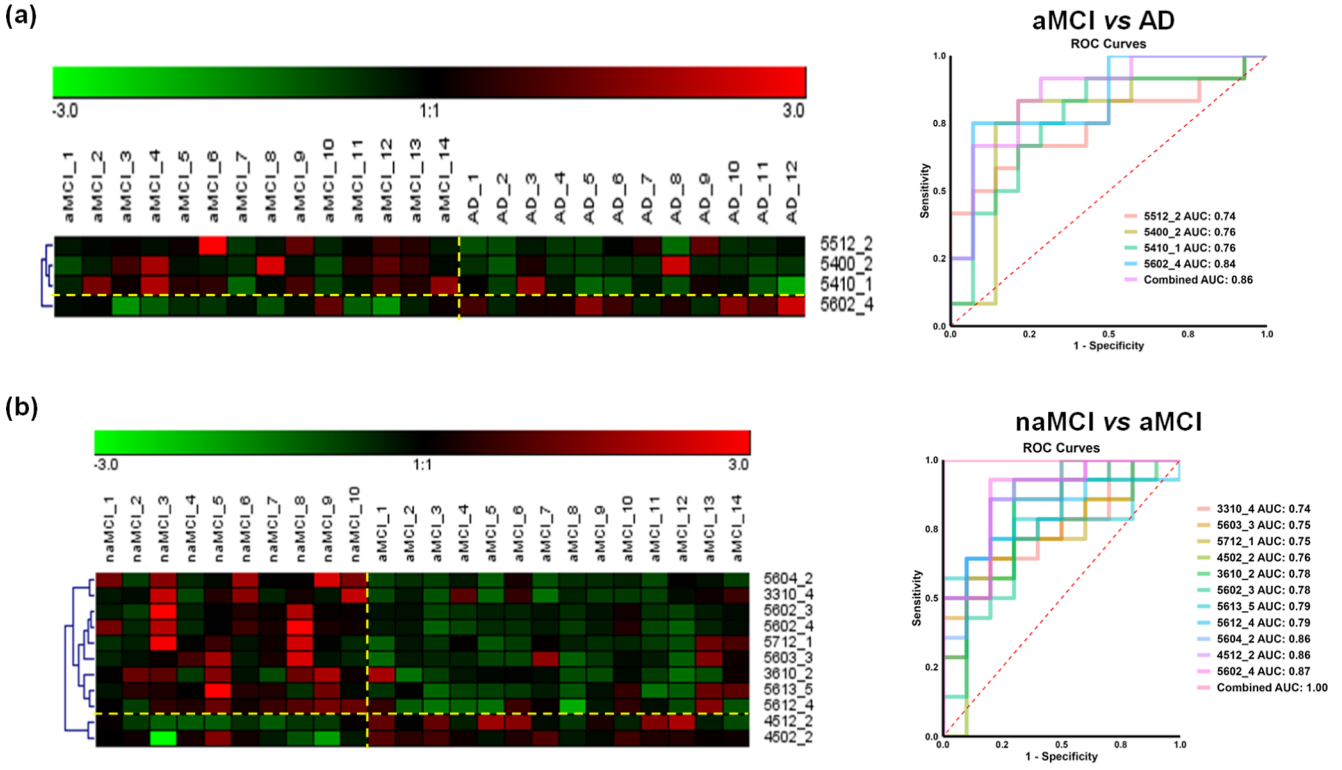


Figure 7. *N*-glycan isomers differentiate MCI subtypes and AD: (a) Heatmap used to visualize the discrepancies in the expression of significant *N*-glycan isomers and ROC curves to determine sensitivity and specificity of candidate biomarkers to monitor progression from aMCI to AD. (b) The heatmap shows the expression of significant *N*-glycan isomers differentiating naMCI from aMCI, and the ROC curve shows their performance.

the leading plasma biomarker for predicting progression to AD and demonstrate diagnostic accuracy equivalent to approved CSF assays.[88−91] A multibiomarker panel is most likely going to provide the most robust disease diagnosis and risk prediction. Indeed, other blood-based biomarkers of neurodegeneration, such as neurofilament light chain (NfL) and glial fibrillary acidic protein (GFAP), detect AD-related neurodegeneration,[92−94] and appear to provide information on disease progression and potential for monitoring treatment effects.[87,95] If validated in future studies, *N*-glycans may provide additional diagnostic or prognostic information and may become an important part of a plasma biomarker AD panel.

Validation of Statistically Significant *N*-Glycan Expression Using LC-PRM-MS

In glycomics studies, discovery-based methods like DDA, as used in this study, help identify potential changes in *N*-glycan expression. To confirm these findings, PRM is employed as it offers more targeted and quantitative validation of these expression changes. PRM enables precise and accurate quantification of specific glycan structures, focusing solely on the glycans of interest. In the present study, we targeted the glycans that exhibited significant expression changes in the full scan analysis to validate our results. We successfully validated the expression changes of six *N*-glycan structures that can differentiate aMCI from AD, including 3511, 5410, 4610, 5521, 6702, and 5711. Additionally, we confirmed the expression of six *N*-glycan structures, including 4310, 4410, 4502, 5603, 6811, and 8601, that can distinguish between MCI subtypes. An example of a representative chromatogram depicting the fragment ions selected for the PRM quantification of *N*-glycan 4502 is shown in Figure S10. The last trace shows the MS/MS spectra containing the fragment selected for this glycan quantitation. Table S5 presents the fold changes in *N*-glycan expression observed in both the full scan and PRM analyses across different comparisons. The average relative abundance of the targeted *N*-glycans is shown in Table S6.

Differential Expression of Isomeric *N*-Glycans

Isomers add significant complexity to glycan structures, contributing to the diversity and functional specificity of these important biomolecules. The presence of isomers in glycans adds a level of diversity that is essential for their biological functions. Understanding the role of isomers in glycan structures has important implications for biology and medicine. Glycans are involved in numerous cellular processes, including cell−cell recognition, signaling, and immune responses.[96,97] Alterations in the expression of these isomeric forms have been correlated to various diseases.[98−100] For example, changes in glycan isomer patterns on cell surfaces are associated with cancer metastasis[101] and immune evasion.[102,103] Glycan isomers have also been shown to be differentially expressed in neurodegenerative diseases including MCI,[30] and narcolepsy type I.[104] Therapeutic strategies targeting glycan isomers, such as glycan-based vaccines and glycosidase inhibitors, are being developed to exploit these differences for disease treatment and prevention. Considering the importance of isomeric glycan expression, we studied their role in the development of AD.

Using the high-temperature nano MGC-LC-MS/MS methodology we previously established,[59,105] combined with the enhanced ionization of permethylated *N*-glycans, we identified and quantified a total of 83 different *N*-glycans consisting of 23 nonisomeric and 60 isomeric *N*-glycans. The 60 isomeric *N*-glycans resulted in 172 *N*-glycans isomers,

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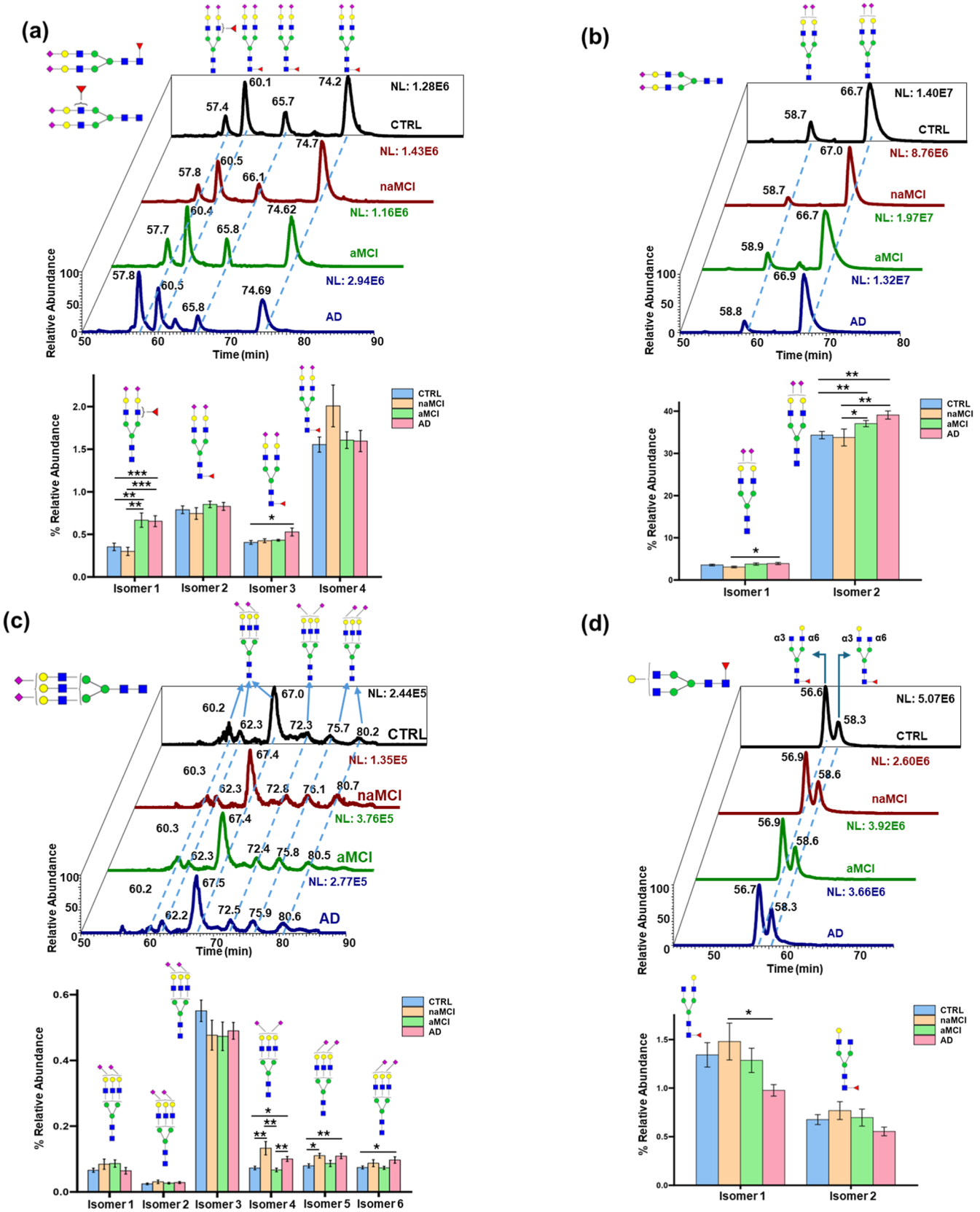


Figure 8. Separation and distribution of *N*-glycan isomers: EICs and bar graphs of statistically significant representative *N*-glycan isomers in the comparison between Control, naMCI, aMCI, and AD. Separation and distribution of (a) $HexNAc_4Hex_5Fuc_1NeuAc_2$ (4512), (b) $HexNAc_4Hex_5NeuAc_2$ (4502), (c) $HexNAc_5Hex_6NeuAc_2$ (5602), and (d) $HexNAc_4Hex_4Fuc_1$ (4410). The error bars represent the SEM (* represents p-value < 0.05, ** represents p-value < 0.01).

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bringing the total number of *N*-glycan structures to 195 for both isomeric and nonisomeric structures. The list of the 195 isomeric *N*-glycan structures identified is shown in Table S7. The average relative abundance of the isomeric structures is provided in Table S8. The identification and comparison of isomeric structures were conducted separately from the compositional analysis because different chromatographic methods were required to achieve optimal separation and resolution of glycan isomers. While compositional profiling was performed using a 150 mm C18 column, isomer-specific analysis required an MGC column operated under high-temperature nano-LC-MS conditions to resolve subtle structural isomers. This separation allowed us to capture fine structural differences not detectable in the compositional analysis. For example, while the compositional analysis identified only the core fucosylated structure, the isomeric analysis revealed four distinct structures for 4512, including both core and branch fucosylated structures.

Most of the isomeric structures contain sialic acid and fucose. We compared the identified *N*-glycans on MGC with the C-18 and found 76 common *N*-glycans between the two methods; 23 were unique to C18 and 8 were unique to MGC (Figure S11). Furthermore, we carried out statistical analysis using the Mann−Whitney *U* test with 95% confidence (p-value < 0.05) followed by FDR correction. We observed 4 isomeric *N*-glycans to be significant in the comparison between the aMCI and AD samples, with 3 downregulated (5512_2, 5400_2, and 5410_1) and 1 upregulated (5602_4) in AD. In this section, the number after the symbol "_" represents the position of the *N*-glycan structures according to their elution. For instance, 5602_4 denotes the fourth isomer of *N*-glycan 5602. The 4 differentially expressed isomeric *N*-glycans were used for an *N*-glycans-specific heatmap to visualize their expression level, as shown in Figure 7a. ROC analysis was also performed, and the result showed an AUC value > 0.7 for all the significant isomeric *N*-glycans and a combined AUC of 0.86 (Figure 7a). Based on the AUC score, the isomeric *N*-glycans are relatively selective and are a good candidate to diagnose and predict progression to AD dementia from aMCI. For the comparison between naMCI and aMCI, 11 isomeric *N*-glycans showed significant changes in expression, 9 of which were downregulated and the remaining 2 were upregulated. ROC analysis showed AUC values > 0.7 for the individual *N*-glycan isomers in this comparison and a combined AUC of 1.

Isomeric comparisons were also conducted between several groups, including CTRL vs naMCI, CTRL vs aMCI, CTRL vs AD, and naMCI vs AD. Figure S12 presents a heat map of the statistically significant *N*-glycan isomers across these comparisons. In the CTRL vs naMCI comparison, four *N*-glycan isomers were significantly altered, all showing upregulation (Figure S12a). In the CTRL vs aMCI comparison, seven isomeric structures were significantly altered, with five downregulated and two upregulated in aMCI compared to the control group (Figure S12b). The comparison between the AD and control groups revealed the most significant changes, with 17 isomeric structures altered of which 11 were downregulated and 6 were upregulated in AD (Figure S12c). Lastly, 9 isomeric structures showed significant changes between naMCI and AD (Figure S12d). The ROC curves demonstrating the potential of these isomeric *N*-glycan structures as biomarkers for early AD detection are displayed in Figure S13.

The excellent separation efficiency and high sensitivity of the high-temperature nanoMGC-LC-MS/MS method enabled efficient profiling of *N*-glycan isomers in serum samples derived from naMCI, aMCI, AD, and healthy control participants. This method allowed for clear resolution of isomers with the same glycan compositions, including highly sialylated structures like 4512 and 5602, fucosylated structures like 4410 and 5410, and neutral structures like 5400 and 3300, leading to improved identification and quantification.

Figure 8 depicts examples of some isomeric separations for *N*-glycans with significant alterations in expression including 4512, 4502, 5602, and 4410. The extracted ion chromatogram of the isomeric *N*-glycan 4512 with 4 distinguishable isomers is shown in Figure 8a. The bar graph below the EICs in this figure shows the percentage relative abundance of the 4 isomers of 4512 and the comparison between the four groups. Isomer 1, identified as a sialyl-Lewis X structure, exhibited the most significant variation across the four groups. This structure was upregulated in aMCI and AD compared to the control and naMCI groups. The observed upregulation of sialyl-Lewis X in AD may indicate enhanced neuroinflammatory signaling through its role in facilitating leukocyte adhesion via selectin binding, potentially contributing to the chronic inflammation seen in AD pathology. The three other isomers are all core fucosylated. Among the core fucosylated structures, only isomer 3 is differentially expressed between control and AD. The tandem MS/MS spectra used for the assignment of these isomers are shown in Figure S14. The diagnostic fragment with m/z 999.5104 was used to distinguish the sialyl-Lewis fragment, while 468.2787 was used to identify core fucosylation as highlighted in the red boxes in Figure S14. In our analysis, we observed two isomers of 4502 as shown in Figure 8b. It has been established that α 2,3-linked sialic acid elutes earlier than α 2,6. For this *N*-glycan, the change in its expression is majorly contributed by its second isomer, which may be α 2,6-linked sialic acid. There is a need for sialidase treatment to validate these glycan isomers. We observed six different isomers of 5602 and the designation was performed according to the NMR data for bovine fetuin (Figure 8c). The first three isomers of 5602, associated with α 2,3-linked sialic acids, showed no significant variation in expression across the groups, while substantial changes were noted in the last three isomers, which predominantly contain α 2,6-linked sialic acids. This suggests that different sialic acid linkages may play distinct roles in disease progression. The significant changes in isomers with 2,6-linked sialic acids indicate a potential closer association with AD. In a study by Nakagawa et al. (2006),[39] overexpression of ST6Gal-1 in N2A cells led to an increase in α 2,6-linked sialylation of APP, which subsequently elevated Aβ levels, a characteristic hallmark of AD. Additionally, we identified another significant isomeric glycan, 4410, which contains a core fucose and terminal galactose. Figure 8d shows galactose-linkage-based isomeric separation from 4410. The order of elution of this monosaccharide site isomer has been shown in our previous study using molecular modeling.[106] We reported that the shorter distance between the core fucose and galactose on α-6 reduces the molecular surface area possibly increasing bond energy. Therefore, the isomer with galactose on the α-6 arm elutes earlier than the α-3 arm. The MS/MS for the identification of the core fucosylation is shown in Figure S15.

Although some of the detected isomers could not be definitively assigned, we still reported the alterations in

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isomeric distribution across the four groups, highlighting the potential roles of glycan isomers in neurodegenerative processes. Exoglycosidases or novel fragmentation techniques that can generate signature fragments may be employed to determine the isomeric structures in future studies. The EICs showing the isomeric distribution of some unassigned glycans are displayed in Figure S16. These include *N*-glycan structures 5410, 3501, 5400, and 3300. Figure S17 presents bar graphs of other significantly altered *N*-glycan isomers across the different groups. The detection of specific *N*-glycan biomarkers associated with neurodegeneration is enhanced by the analysis of *N*-glycan isomers, as it reveals glycosylation alterations that may not be evident when only total *N*-glycan levels are analyzed.

Correlation of Serum *N*-Glycans with Established Alzheimer's Disease Biomarkers

To support the potential of serum-based *N*-glycan biomarkers in AD, we conducted a correlation analysis between the differentially expressed *N*-glycans and key AD biomarkers, including pT181, pT217, total tau (tau), aBeta42, aBeta40, aBeta42/40 ratio, NfL, and GFAP. Notably, *N*-glycan 5521, which emerged as a strong candidate for distinguishing aMCI from AD (AUC = 0.99), demonstrated consistent positive correlations with tau-related biomarkers (tau, pT181, and pT217) and GFAP in AD, as illustrated in the correlation heatmap shown in Figure S18. Interestingly, the *N*-glycan 5521 shows an inverse correlation with tau, pT181, pT217, and GFAP in aMCI, supporting its biological relevance in differentiating aMCI from AD Figure S19. Several other candidate *N*-glycan biomarkers identified in our study correlate strongly with AD, as depicted in Figure S18. Glycans including 5410, 4610, 3610, 3501, and 5411, have a strong inverse correlation with the established AD biomarkers. Other glycans like 5604,6702, and 4502 correlate positively with AD biomarkers.

CONCLUSION

This study has provided a detailed exploration of *N*-glycan profiles associated with AD and MCI, specifically examining both isomeric and nonisomeric *N*-glycans using LC−MS/MS and advanced glycomics approaches. We successfully identified 99 unique *N*-glycans in nonisomeric analysis across different stages of cognitive decline. Glycan profiling was conducted on serum samples from participants with normal cognition, naMCI, aMCI, and AD, revealing specific glycan structures that may serve as biomarkers for early diagnosis and disease progression monitoring. Significant changes in glycan expression, including sialylation and fucosylation, were observed, with sialylation showing an increase in AD patients while fucosylation decreased significantly, potentially linking these modifications to AD pathology. The alterations in sialylation and fucosylation offer insights into potential therapeutic targets. Modulating these glycosylation pathways may help to slow or alter the progression of AD. Notably, we observed that changes in the expression of the PRM validated *N*-glycans 3511, 5410, 4610, 5521, 6702, and 5711 can distinguish aMCI from AD dementia while *N*-glycans 4310, 4410, 4502, 5603, 6811, and 8601 can differentiate naMCI from aMCI. The isomeric profiling of *N*-glycans further revealed critical insights, identifying several isomers that differentiate between MCI subtypes and AD. These findings emphasize the importance of *N*-glycan structures as potential diagnostic tools and therapeutic targets for AD. The glycan structures identified in this study have the potential to serve as early diagnostic biomarkers for AD and its prodromal stages. The specific glycan signatures and their isomeric forms could be developed into blood-based tests for more accessible and less invasive diagnosis and prediction of AD progression. Future research involving a larger sample cohort, encompassing diverse ethnic backgrounds, ages, and genders, with longitudinal samples throughout the disease course will be necessary to confirm the identified glycan biomarkers and clarify the precise role of glycans in AD and its progression.

ASSOCIATED CONTENT

Data Availability Statement

The raw data is available on GlycoPost with accession number: GPST000498.[107]

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jproteome.5c00018.

MCI and AD Glycomics Supporting Information. Figure S1: Combined Unsupervised Principal Component Analysis of *N*-glycans derived from Control, naMCI, aMCI, and AD groups at 95% confidence level; Figure S2: Examples of MS/MS spectra of some *N*-glycans showing spectra for *N*-glycan assignment, including (a) $HexNAc_4Hex_5Fuc_1NeuAc_2$ (4512) (b) $HexNAc_5Hex_6Fuc_1NeuAc_3$ (5613) (c) $HexNAc_5Hex_5Fuc_1NeuAc_2$ (5512), (d) $HexNAc_5Hex_6NeuAc_3$ (5603) (e) $HexNAc_6Hex_7NeuAc_3$ (6703) (f) $HexNAc_8Hex_6NeuAc_1$ (8601) (g) $HexNAc_3Hex_5NeuAc_1$ (3501) (h) $HexNAc_5Hex_5Fuc_1NeuAc_1$ (5511); Figure S3: Bar graphs showing the sialylation and fucosylation types influencing the changes in these glycan types across the investigated groups. Bar graph showing (a) sialylation types according to the number of sialic acids attached, (b) fucosylation types according to the number of fucose moieties attached, and (c) distribution of core fucosylation across groups. Control (n = 28), naMCI (n =10), aMCI (n = 14), and AD (n = 12). (* represents p-value < 0.05, ** represents p-value < 0.01); Figure S4: MS/MS spectra of representative core fucosylated *N*-glycans including (a) $HexNAc_4Hex_3Fuc_1$ (4310), (b) $HexNAc_4Hex_4Fuc_1$ (4410), and (c) $HexNAc_5Hex_3Fuc_1$ (5310); Figure S5: Heatmap representation of statistically significant *N*-glycans in the comparison between (a) control and naMCI, (b) control and aMCI, (c) control and AD, and (d) naMCI and AD; Figure S6: volcano plots showing the statistical distribution of all *N*-glycans in the comparison between (a) control and naMCI, (b) control and aMCI, (c) control and AD, and (d) naMCI and AD; Figure S7: box plots of statistically significant *N*-glycans in any of the four comparisons: control *vs* naMCI, control vs aMCI, control vs AD, and naMCI vs AD; Figure S8: ROC analysis showing the performance of significant *N*-glycans in the comparison between (a) control and naMCI, (b) control and aMCI, (c) control and AD, and (d) naMCI and AD; Figure S9: Venn plot showing similarities and disparities in the expression of significant *N*-glycans in the comparison between (a) control and other groups, (b) naMCI and other groups; Figure S10: representative chromatograms of *N*-glycan $HexNAc_4Hex_5NeuAc_2$ (4502) in control,

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naMCI, aMCI, and AD using targeted PRM analysis; Figure S11: Venn plot depicting the common and distinct *N*-glycans identified in the serum samples using C18 and MGC LC−MS/MS; Figure S12: Heatmap representation of statistically significant *N*-glycan isomers in the comparison between (a) control and naMCI, (b) control and aMCI, (c) control and AD, and (d) naMCI and AD; Figure S13: ROC analysis showing the performance of significant *N*-glycan isomers in the comparison between (a) control and naMCI, (b) control and aMCI, (c) control and AD, and (d) naMCI and AD; Figure S14: MS/MS spectra showing the diagnostic fragments used to assign the glycan isomers for $HexNAc_4Hex_5Fuc_1NeuAc_2$ (4512); Figure S15: MS/MS spectra showing the diagnostic fragments used to assign the glycan isomers for $HexNAc_4Hex_4Fuc_1$ (4410). The galactose-based arm branching is assigned using the retention time pattern of the glycan structures; Figure S16: EICs and box plots of statistically significant *N*-glycan isomers in the comparison between the four groups. Separation and distribution of (a) $HexNAc_5Hex_4Fuc_1$ (5410), (b) $HexNAc_3Hex_5NeuAc_1$ (3501), (c) $HexNAc_5Hex_4$ (5400), (d) $HexNAc_3Hex_3$ (3300); Figure S17: bar graphs showing the distribution of the remaining significant *N*-glycan isomers, including sialylated, sialofucosylated, and fucosylated *N*-glycan isomers; Figure S18: correlation heatmap showing the relationships between serum *N*-glycan biomarkers and established AD biomarkers; Figure S19: correlation heatmap showing the relationships between serum *N*-glycan biomarkers (four-digit codes) and established AD biomarkers in aMCI. Table S1: transition ion fragments for the targeted validation of statistically significant *N*-Glycans in the full scan analysis; Table S2: list of 99 *N*-glycans including their nomenclature, cartoons, theoretical *m*/*z*, and corresponding mass accuracy; Table S3: List of unique *N*-Glycan compositions identified in the individual groups; Table S4: Relative abundance of the identified *N*-glycans. A four-digit *N*-glycan nomenclature was used in the following order: *N*-acetylglucosamine, hexose, fucose, and *N*-acetylneuraminic acid (GlcNAc, Hex, Fuc, NeuAc); Table S5: statistically significant *N*-Glycans validated by targeted LC-PRM-MS showing fold change (FC) in full scan and PRM; Table S6: average relative abundance of the targeted LC-PRM-MS analysis of the statistically significant *N*-glycans; Table S8: relative abundance of the identified *N*-glycan isomers. A four-digit *N*-glycan nomenclature was used in the following order: *N*-acetylglucosamine, hexose, fucose, and *N*-acetylneuraminic acid (GlcNAc, Hex, Fuc, NeuAc) (PDF)

Figure S2: Examples of MS/MS spectra of some *N*-glycans showing spectra for *N*-glycan assignment (XLSX)

Figure S3: Bar graphs showing the sialylation and fucosylation types influencing the changes in these glycan types across the investigated groups (XLSX)

Table S7: List of 195 isomeric *N*-glycan structures including their nomenclature, cartoons, theoretical *m*/*z*, and corresponding mass accuracy (XLSX)

AUTHOR INFORMATION

Corresponding Author

Yehia Mechref − *Department of Chemistry and Biochemistry, Texas Tech University, Lubbock, Texas 79409, United States;* orcid.org/0000-0002-6661-6073; Phone: 806-742-3059; Email: yehia.mechref@ttu.edu; Fax: 806-742-1289

Authors

Sherifdeen Onigbinde − *Department of Chemistry and Biochemistry, Texas Tech University, Lubbock, Texas 79409, United States*

Joy Solomon − *Department of Chemistry and Biochemistry, Texas Tech University, Lubbock, Texas 79409, United States*

Cristian D. Gutierrez-Reyes − *Department of Chemistry and Biochemistry, Texas Tech University, Lubbock, Texas 79409, United States*

Oluwatosin Daramola − *Department of Chemistry and Biochemistry, Texas Tech University, Lubbock, Texas 79409, United States*

Mojibola Fowowe − *Department of Chemistry and Biochemistry, Texas Tech University, Lubbock, Texas 79409, United States*

Moyinoluwa Adeniyi − *Department of Chemistry and Biochemistry, Texas Tech University, Lubbock, Texas 79409, United States*

Kelly N. DuBois − *Department of Translational Science and Molecular Medicine, College of Human Medicine, Michigan State University, Grand Rapids, Michigan 49503, United States; Michigan Alzheimer's Disease Research Center, University of Michigan, Ann Arbor, Michigan 48103, United States*

Kelly M. Bakulski − *Department of Epidemiology, School of Public Health, University of Michigan, Ann Arbor, Michigan 48103, United States*

Nicholas M. Kanaan − *Department of Translational Science and Molecular Medicine, College of Human Medicine, Michigan State University, Grand Rapids, Michigan 49503, United States; Michigan Alzheimer's Disease Research Center, University of Michigan, Ann Arbor, Michigan 48103, United States*

David M. Lubman − *Department of Surgery, The University of Michigan, Ann Arbor, Michigan 48109, United States;* orcid.org/0000-0001-7731-0232

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.jproteome.5c00018

Author Contributions

Conceptualization, Yehia Mechref, David M. Lubman, and Sherifdeen Onigbinde; project administration, supervision, and funding acquisition, Yehia Mechref, David M. Lubman and Nicholas M. Kanaan; methodology and formal analysis, Sherifdeen Onigbinde, Joy Solomon, Cristian D. Gutierrez-Reyes, and Oluwatosin Daramola; data curation, Sherifdeen Onigbinde, Joy Solomon, Mojibola Fowowe, Moyinoluwa Adeniyi, and Cristian D. Gutierrez-Reyes; visualization, Sherifdeen Onigbinde, Joy Solomon, Moyinoluwa Adeniyi, and Oluwatosin Daramola; writing original draft preparation, Sherifdeen Onigbinde, and Oluwatosin Daramola writing review and editing, Sherifdeen Onigbinde, Kelly N. Dubois, Nicholas M. Kanaan, Kelly M. Bakulski, Joy Solomon, Oluwatosin Daramola, Moyinoluwa Adeniyi, Cristian D.

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Gutierrez-Reyes, David M. Lubman, and Yehia Mechref; sample design and acquisition, Kelly M. Bakulski, Kelly N. Dubois, Nicholas M. Kanaan, and David M. Lubman. All authors have approved the final version of the manuscript.

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Notes

The authors declare no competing financial interest.

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